There are three types of Species:

a. **Endemic species** - is one whose habitat is restricted only to a particular area because of which it is often endangered. It differs from "indigenous," or "native," which although it occurs naturally in an area, is also found in other areas.

b. **Exotic Species** - is any species intentionally or accidentally transported and released by man into an environment outside its original range. These are often the most severe agents of habitat alteration and degradation, and a major cause of the continuing loss of biological diversity throughout the world.

c. **Cosmopolitan Species** – It is a species that is found to be distributed over most regions of the earth example: cats, dogs, human beings. The killer whale is considered as the most cosmopolitan species in the world.

C. Ecosystem diversity refers to the variety of life forms in a prescribed ecosystem. Ecosystems may be both terrestrial and aquatic. Distinctive terrestrial ecosystems include forests, grasslands, deserts, etc. while aquatic ecosystems are rivers, lakes, oceans etc.

In understanding biodiversity, the most common question that arises in our mind is how many different plant and animal species are there on earth? There can be no definite answer to this question. At present the conservation scientists have identified over 8.7 million species worldwide. Of this only about 2 million are known to us ranging from microorganisms to giant mammals and reptiles. New species are being discovered while many species are also disappearing from the face of the earth.

7.5.1. Biodiversity hotspots

Areas that are rich in species diversity are called as "Hotspots". The hottest spots for species diversity are the tropical rainforests. Tropical rainforests comprise of only 7% of all land on earth, yet are home to nearly 50% of all the species on Earth! India is among the World's 17 nations that are exceptionally rich in species diversity.

The British biologist Norman Myers coined the term 'biodiversity hotspot' in 1988. According to him, a biodiversity hotspot is a biogeographic region characterised both by exceptional levels of plant endemism and by serious levels of habitat loss. **Conservation International** (CI) adopted Myers concept of 'hotspots' and it made an extensive global study of hotspots in 1999. According to CI, to qualify as a hotspot a region must meet two strict criteria: (i) It must contain at least 1,500 species of endemic plants, and (ii) It must have lost at least 70% of its original habitat. In 1999, CI's book 'Hotspots: Earth's Biologically Richest and Most Endangered Terrestrial Ecoregions', identified 34 biodiversity hotspots in the different countries of the world.

Currently there are 34 biodiversity hotspots that have been identified and, most of them occur in **tropical forests** (Figure 7.10). They represent just 2.3% of Earth's land surface, but between them they contain around 50% of the world's

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